

FRANKLINE MISANGO OYOLO

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Education

The University of Hong Kong

Aug 2022 – Jun 2027

Bachelor of Engineering in Data Science and Engineering— GPA: 3.70/4.3

Hong Kong SAR

- **Relevant courses:** Data Structures & Algorithms, Machine Learning, Stochastic Calculus, Probability & Statistics

Professional Experience

The Swiss Federal Institute of Technology

Jul 2025 – August 2025

PhD Research Assistant

Switzerland

- Developed graph-augmented multi-factor panel model using DCRNN framework on 50+ TB Swiss transaction data, achieving out-of-sample $R^2=0.32$ and $MAPE=6.4\%$ for expenditure flows within cities through residual graph networks
- Designed regime-switching detection system combining HMM state classification with GARCH volatility overlays, demonstrating 21-day median lead time vs macroeconomic releases. Integrated factor optimizer reducing active risk by 60 basis points
- Analyzed 1 petabyte historical panel to develop canton-level discretionary demand factor using graph neural network message passing, improving allocation model IR by +0.28 through spatial dependency modeling

BASF SE

Jan 2025 – Jun 2025

Machine Learning Researcher

Hong Kong SAR

- Modeled trade policy shocks using panel difference-in-differences with local projection estimators, generating 50k Monte Carlo paths that reduced DSO volatility by 42% and improved cash flow forecast RMSE by 18%
- Optimized global supplier networks through CVaR-constrained Monte Carlo simulation, achieving 25bps landed cost variance while releasing \$2.3M working capital via tariff pass-through minimization
- Developed 300+ SHAP-interpretable automated valuation models ($R^2=0.92$) decomposing enterprise value drivers (volume, ASP, FX, WACC), guiding semiconductor sales strategy with projected 14% EBIT uplift

Manulife Financial Corporation

Jun 2024 – Aug 2024

Data Governance Summer Analyst

Hong Kong SAR

- Built metadata enrichment engine processing 2.3M+ Azure Purview assets ($F1\text{-score}=87.2\%$), increasing tag coverage from 4% to 65% and actuarial model feature completeness by 18 percentage points
- Implemented column-level lineage tracking using Snowflake and GitOps, mapping premium/claims data to IFRS 17 calculation layers and reducing model onboarding time by 40%
- Automated 300+ data quality checks with Great Expectations framework, reducing critical error rate by 62% while improving new business margin screening precision to 92%

Quantitative Research and Development

Cross-Regime Analysis of Leveraged Portfolio Strategies

HKU Trading Group Published

- Formalized regime-switching portfolio optimization via HJB equations and risk parity constraints ($\alpha = 0.92$), achieving 46.2% annual returns (2017-2021) with 22% lower volatility than SPY, identifying 3 distinct regimes through Markov switching analysis
- Modelled volatility decay mitigation framework using GARCH(1,1)-optimized ERC weights (Adjusted Sigma = 1.2), reducing leveraged ETF decay by 38% compared to naive 3x allocation in backtests
- Optimized correlations across 12 different financial instruments using dynamic correlation adjustments over time, reducing the maximum potential loss by 18.2%, achieving a Sharpe ratio of 0.82 in stress tests, and improving the stability and performance of the portfolio in volatile market conditions.

Market Making Through Jump Diffusion HJB Equations with Order Flow Toxicity Tracking

Peer Review

- Enhanced HJB equations with jump diffusion modeling ($\lambda = 3.2$ jumps/day) achieving 1.48 Sharpe ratio in BTC/USD backtests, reducing inventory risk by 62% vs traditional quoting strategies
- Implemented finite difference methods on 201x201 grid with 0.89ms latency using CUDA-optimized kernels (136x speedup vs CPU), maintaining 97.4% theoretical PnL efficiency after transaction costs
- Developed flow toxicity tracking system reducing adverse selection by 41% through real-time order imbalance detection (Squared R = 0.79) vs actual toxic flow events)

FeenQR — Quant Research Platform

C# — 2025-Present

- Designed and launched a real-time quantitative research platform with seamless Bloomberg B-Pipe integration, achieving **92% strategy accuracy** and a Sharpe ratio of 3.8 in simulated trading
- Constructed a distributed backtesting engine capable of processing **2.4TB of market data per hour** on an 8-GPU A100 cluster, enabling rapid exploration and validation of trading strategies
- Integrated large language models to power an AI-driven alpha generation module, delivering **17% CAGR** in simulated trading performance

AlgoForge Backtester — Algorithmic Trading Backtesting Ecosystem

Python/C# — 2024-2025

- Architected AI-powered strategy generator (RAGENTIC) converting natural language to production-grade C# algorithms with **94% compilation success rate** on first attempt
- Built distributed data pipeline processing **1.8TB/day** across 12+ sources (Alpaca/Binance/Polygon) with automated LEAN format conversion and quality checks
- Implemented QuantConnect integration enabling backtesting of AI-generated strategies with **3.2x speed improvement** over vanilla LEAN CLI through optimized slippage modeling
- Developed ML-powered data quality analyzer achieving **99.7% integrity validation** across 15+ market data formats (Tick/CSV/Parquet/HDF5)